Mathematics for Machine Learning

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Probability

Statistics

Convex Optimization

Probability

Probability is about measuring **uncertainty**.

Interpretation of probability

- **Frequentist**: probability is relative frequency in the long run of an event (assuming it can happen many times).
- **Bayesian**: probability is the degree of belief. It quantifies our ignorance about something. Tt is more about information than repeated trials

Probability measure

A **probability space** is a triplet $(\Omega, \mathcal{F}, P)$ consisting:

- **Sample space**: the set of all possible outcomes.
- **Events set**: a σ-**algebra** on the sample space. It is a non-empty collection of subsets of Ω closed under complement and countable unions.
- **Probability measure**: a set function $P : \mathcal{F} \to [0, 1]$ such that:
  - $P(\varnothing) = 0$,
  - $P(\Omega) = 1$,
  - $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$ for pairwise disjoint events ($A_i \cap A_j = \varnothing$ for $i \neq j$)

Rules of probability

Given events $A, B \in \mathcal{F}$ with $P(B) > 0$, the **conditional probability** of A given B is

$$P(A|B) = P(A \cap B)/P(B)$$

Rules of probability

- **Bayes rule**: $P(A|B)P(B) = P(B|A)P(A)$
- **Sum rule**: Given (B_n) a countable partition of Ω, $P(A) = \sum_n P(A, B_n) = \sum_n P(A|B_n)P(B_n)$.
- **Product rule**: Given $A_1 ... A_n \in \mathcal{F}$ $P(A_1, ..., A_n) = P(A_1)P(A_2|A_1)...P(A_n|A_1, ..., A_{n-1})$

Limit theorems

Law of large numbers

The sample mean of a sequence of i.i.d random variables with finite variance converges almost surely to the expected value.

Central limit theorem

The standardized sample mean (or sum) of a sequence of i.i.d random variables with finite and non-zero variance converges in distribution to the standard normal.

Limit theorems

Law of large numbers

Given a sequence off i.i.d random variables $(X_n)_{n\in\mathbb{N}}$ with finite variance,

$$\overline{X} = \frac{1}{N}\sum_{i=1}^{N} X_i \overset{a.s.}{\to} E[X_i]$$

Central limit theorem

Given a sequence off i.i.d random variables $(X_n)_{n\in\mathbb{N}}$ with mean μ and variance $0 < \sigma^2 < \infty$, (note that the mean of $E[\overline{X}] = \mu$ and $V[\overline{X}] = \frac{\sigma^2}{N}$),

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \overset{P}{\to} \mathcal{N}(0,1)$$

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Inferential Statistics

- **Statistical inference** (statistical learning/modeling/estimation) is the process of using a (random) sample to infer the distribution (or some properties of the distribution) that generated the sample.
- We resort to statistics (and not just plain inference) because the data is prone to random variations (noice, uncertainty).

Parametric inference

- A **statistical model** is a family of probability distributions $(p_\theta)_{\theta \in \Theta}$. A model is used for prediction and/or explanation.
- A **parametric model** is a model that can be parametrized by a finite number of parameters (not a function of the data).

Goal

The data is a sequence $x_1, ..., x_n$ of n i.i.d random variables from some distribution p parametrized by θ. The goal is to estimate the fixed but unknown quantity θ.

Estimator

- An **estimator** $\hat{\theta}_n$ of θ is a function of the data: $\hat{\theta}_n = g(x_1, ..., x_n)$. It is a random variable. Its distribution is called the **sampling distribution**.

Properties of an estimator

- The **bias** of an estimator is $\text{bias}(\hat{\theta}_n) = E[\hat{\theta}_n] - \theta$. An estimator is unbiased if the bias is zero or $E[\theta_n] = \theta$.
- An estimator is **consistent** if it converges in probability to the true value of the parameter: $\hat{\theta}_n \xrightarrow{P} \theta^*$.
- The quadratic risk or mean squared error of an estimator is $\text{MSE} = E[(\hat{\theta}_n - \theta)^2]$. It is used to evaluate the quality of the estimator. It can be written as $\text{MSE} = \text{bias}^2(\hat{\theta}_n) + V(\hat{\theta}_n)$.

Maximum Likelihood Estimator

We assume the observations are i.i.d from some distribution p.

- The **likelihood** function is the (joint) probability of the data:

$$L(\theta) = p(\mathcal{D}; \theta) = \prod_{i=1}^{n} p(x_i; \theta)$$

- The **Maximum likelihood estimate** is the value of θ that maximizes the likelihood function:

$$\hat{\theta}_n = \text{argmax}_\theta L(\theta)$$

- It is consistent, equivarient, asymptotically Normal, and asymptotically efficient.

In practice, we instead minimize the negative log-likelihood.

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Outline

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Convex Optimization
- Convex functions
- Convex optimization problems
- Unconstrained Minimization

Convex functions

- $f : \mathbb{R}^n \to \mathbb{R}$ is convex if **dom** f is convex and

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y)$$

  for all $x, y \in$ **dom** f, $0 \leq \theta \leq 1$
- f is concave if $-f$ is convex.

Examples on $\mathbb{R}$

Convex

- affine : $ax + b$ on $\mathbb{R}$, $a, b \in$ on $\mathbb{R}$
- exponential : e^{ax} on $\mathbb{R}$, $a \in \mathbb{R}$
- powers : x^α on $\mathbb{R}_{++}$, for $\alpha \geq 1$ or $\alpha \leq 0$

Concave

- affine
- powers : x^α on on $\mathbb{R}_{++}$, for $0 \leq \alpha \leq 1$
- logarithm : $\log(x)$ on $\mathbb{R}_{++}$

Examples on $\mathbb{R}^n$

- Affine (linear) functions are both convex and concave

$$f(x) = a^\top x + b$$

- Norms are convex

$$||x||_p = (\sum_{i=1}^{n} |x|^p)^{1/p}$$

First-order condition

f is differentiable if the gradient $\nabla f(x) = \left(\frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_1}, ..., \frac{\partial f(x)}{\partial x_n} \right)$ exists at each $x \in$ **dom** f.

First-order condition

f is differentiable with a convex domain. f is convex iff

$$f(y) \geq f(x) + \nabla f(x)^\top (x - y)$$

for all $x, y \in$ **dom** f

Second-order condition

f is twice differentiable if the Hessian $\nabla^2 f(x)$,

$$(\nabla^2 f(x))_{i,j} = \frac{\partial f}{\partial x_i \partial x_j} \; i, j = 1, ..., n$$

exists at each $x \in$ **dom** f

Second-order condition

f is twice differentiable with a convex domain. f is convex iff its Hessian is positive semidefinite.

$$\nabla^2 f(x) \succeq 0, \text{for all} x \in \textbf{dom } f$$

Convexity in practice

Practical method for establishing convexity of a function :

1. Verify definition
2. For twice differentiable functions, show $\nabla^2 f(x) \succeq 0$
3. Show that f is obtained from simple convex functions by operations that preserve convexity

Operations preserving convexity

- nonnegative scaling : f convex, $\alpha \geq 0 \implies \alpha f$ convex
- sum : f, g convex $\implies f + g$ convex
- affine composition : f convex $\implies f(Ax + b)$ convex
- pointwise maximum : $f_1, ..., f_k$ convex $\implies max_i f_i(x)$ convex
- composition : h convex increasing, f convex $\implies h(f(x))$ convex

similar rules for concave functions (there are more)

The only rule you need to know

$h(f_1(x), ..., f_k(x))$ is convex if h convex and

- h increasing in each argument, and f_i are convex, or
- h decreasing in each argument, and f_i are concave, or
- f_i are affine

similar rules for concave (just swap convex and concave)

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Optimization problem in standard form

$$\begin{aligned} \text{minimize } & f_0(x) \\ \text{s.t } & f_i(x) \leq 0,\ i = 1, ..., m \\ & g_i(x) = 0,\ i = 1, ..., p \end{aligned}$$

- $x \in \mathbb{R}^n$ is the optimization variable
- $f_0 : \mathbb{R}^n \to \mathbb{R}$ is the **objective or cost function**
- $f_i : \mathbb{R}^n \to \mathbb{R},\ i = 1, ..., m$ are the **inequality constraint** functions
- $h_i : \mathbb{R}^n \to \mathbb{R},\ i = 1, ..., p$ are the **equality constraint** functions

Feasibility

- The **domain** (also called **implicit constraint**) is the set of points for which the objective and all constraint functions are defined

$$\mathcal{D} = \bigcap_{i=0}^{m} \mathbf{dom}\ f_i \cap \bigcap_{i=1}^{p} \mathbf{dom}\ h_i$$

- A problem is **unconstrained** if it has no explicit contraints.
- A point $x \in \mathcal{D}$ is **feasible** if it satisfies the constraints.
- The set of all feasible points is called the **feasible set** or the **constraint set**.
- The problem is said to be **feasible** if there exists at least one feasible point, and **infeasible** otherwise.

Optimal value and points

- The **optimal value** $p^\star$ of the problem is

$$p^\star = \inf\{f_0(x) \mid f_i(x) \leq 0,\ i = 1, ..., m,\ g_i(x) = 0,\ i = 1, ..., p\}$$

  - $p^\star = \infty$ if the problem is infeasible
  - $p^\star = -\infty$ if the problem is unbounded below
- $x^\star$ is an **optimal point** (or solves the problem) if $x^\star$ is feasible and $f_0(x^\star) = p^\star$
- The set of optimal points is the **optimal set**

$$X_{opt} = \inf\{x \mid f_i(x) \leq 0,\ i = 1, ..., m,\ g_i(x) = 0,\ i = 1, ..., p,\ f_0(x^\star) = p^\star\}$$

- If there exists an optimal point for the problem , we say the optimal value is **attained** or **achieved**, and the problem is **solvable**.
- A point x is **locally optimal** if it is feasible and there is an $R > 0$ such that

$$f(x) = \inf\{f_0(z) \mid f_i(z) \leq 0,\ i = 1, ..., m,\ g_i(z) = 0,\ i = 1, ..., p,\ ||z - x||_2 < R)\}$$

Convex optimization problems

A **convex optimization problem** is of the form

$$\begin{aligned} &\text{minimize } f_0(x) \\ &\quad \text{s.t } f_i(x) \leq 0,\ i = 1, ..., m \\ &\quad\quad a_i^\top x = b_i,\ i = 1, ..., p \text{ (or } Ax = b) \end{aligned}$$

where $f_i,\ i = 0, ..., m$ are convex functions. Compared to standard problems, the convex problem has three more requirements:

- the objective function must be convex,
- the inequality constraint functions must be convex,
- the equality constraint functions must be affine

The feasible set of a convex optimization problem is convex.

Local and global optima

Any locally optimal point of a convex problem is also (globally) optimal.

Equivalent convex problems

We call two problems **equivalent** if from a solution of one, a solution of the other is readily found, and vice versa.
There are general transformations that yield equivalent problems including:

- change of variables
- transformation of objective and constraint functions
- slack variables
- eliminating or introducing equality constraints
- epigraph problem form

Linear program (LP)

$$\begin{aligned} \text{minimize } & c^\top x + d \\ \text{s.t } & Gx \preceq 0 \\ & Ax = b \end{aligned}$$

where $G \in \mathbb{R}^{m \times n}$ and $A \in \mathbb{R}^{p \times n}$

- It is a convex problem with affine objective and constraint functions.
- The feasible set is a polyhedron.

Quadratic program (QP)

$$\begin{aligned} \text{minimize } & (1/2)x^\top Px + q^\top x + r \\ \text{s.t } & Gx \preceq 0 \\ & Ax = b \end{aligned}$$

where $P \in S_+^n$ (positive semi-definite), $G \in \mathbb{R}^{m \times n}$ and $A \in \mathbb{R}^{p \times n}$

- we minimize a convex quadratic function over a polyhedron

Quadratically constrained quadratic program (QCQP)

$$\begin{aligned} \text{minimize } & (1/2)x^\top P_0 x + q_0^\top x + r_0 \\ \text{s.t } & (1/2)x^\top P_i x + q_i^\top x + r_i \leq 0,\ i = 1, ..., m \\ & Ax = b \end{aligned}$$

where $P_i \in S_+^n$

- the objective and the inequality constraint functions are (convex) quadratic
- if $P_i \succ 0$, we minimize a convex quadratic function over a feasible region that is the intersection of ellipsoids

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Unconstrained minimization problems

$$\text{minimize } f(x)$$

where $f : \mathbb{R}^N \to \mathbb{R}$ is convex and twice differentiable. We assume that the optimal value $p^\star = \inf_x f(x)$ is attained (and finite).

Optimality condition

f is differentiable and convex, a point $x^\star$ is optimal iff $\nabla f(x^\star) = 0$.

Sometimes, we can analytically solve the optimality equation but usually the problem must be solved by an **iterative algorithm**.

Descent methods

We want to produce a minimizing sequence $x^{(k)}$, $k = 1, ...,$ where $x^{(k+1)} = x^{(k)} + t^{(k)}\Delta x^{(k)}$ with $f(x^{(k+1)}) < f(x^{(k)})$

- $\Delta x^{(k)}$ is the **search direction**
- $t^{(k)} > 0$ is the **step size**

Descent direction

$\Delta x^{(k)}$ is a **descent direction** if $\nabla f(x^{(k)})^\top \Delta x^{(k)} < 0$ (from convexity).

Algorithm 1: General descent method

Start at a point $x \in$ **dom** f

repeat

  Determine a descent direction Δx

  Choose a step size $t > 0$

  Update $x = x + t\Delta x$

until *stopping criterion is satisfied*;

Gradient descent

It is a descent method with search direction $\Delta x = -\nabla f(x)$.

Algorithm 2: Gradient descent

Start at a point $x \in$ **dom** f

repeat

- $\Delta x = -\nabla f(x)$
- Choose a step size $t > 0$
- Update $x = x + t\Delta x$

until *stopping criterion is satisfied*;

The stopping criterion is usually of the form $||\nabla f(x)||_2 < \epsilon$.

Newton's method

It is a descent method with search direction $\Delta x = -(\nabla^2 f(x))^{-1} \nabla f(x)$.

We can say that $v = \Delta_x$

- minimizes the second order approximation

$$f(x+v) \approx f(x) + \nabla f(x)^\top v + \frac{1}{2} v^\top \nabla^2 f(x) v$$

- or equivalently solves the linearized optimality condition

$$\nabla f(x+v) \approx \nabla f(x) + \nabla^2 f(x) v = 0$$

More

Explore further

- Steepest descent methods (generalizes gradient descent)
- Quasi-Newton methods (like BFGS, L-BFGS)
- Convergence analysis
- Self-concordance
- Gradient-free optimization